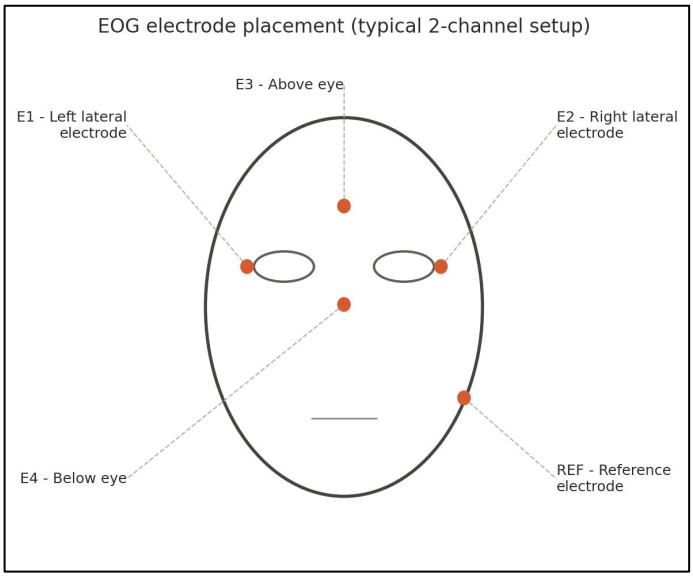
Assistive EOG Blink Switch

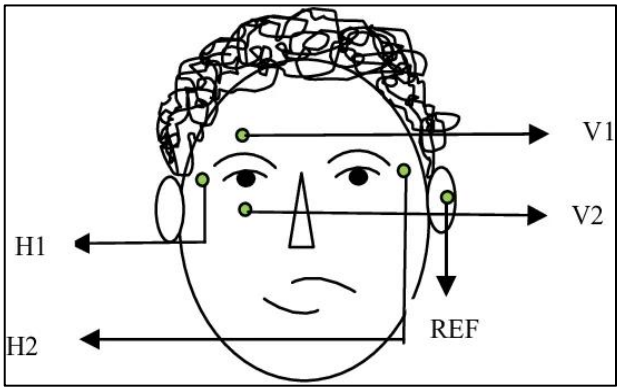
Giving control back through the eyes: communication, room automation, and emergency alerting for people with limited mobility

1. Introduction and motivation(inspiration/ideas)

For people living with conditions such as ALS, high-level spinal cord injury, or severe cerebral palsy, the muscles that move the eyes are often the last ones to remain under voluntary control - even after speech and hand movement are lost. That makes the eyes the last remaining channel a person has to act on the world. This project turns that one channel into a way to communicate, control a room, and call for help in an emergency.

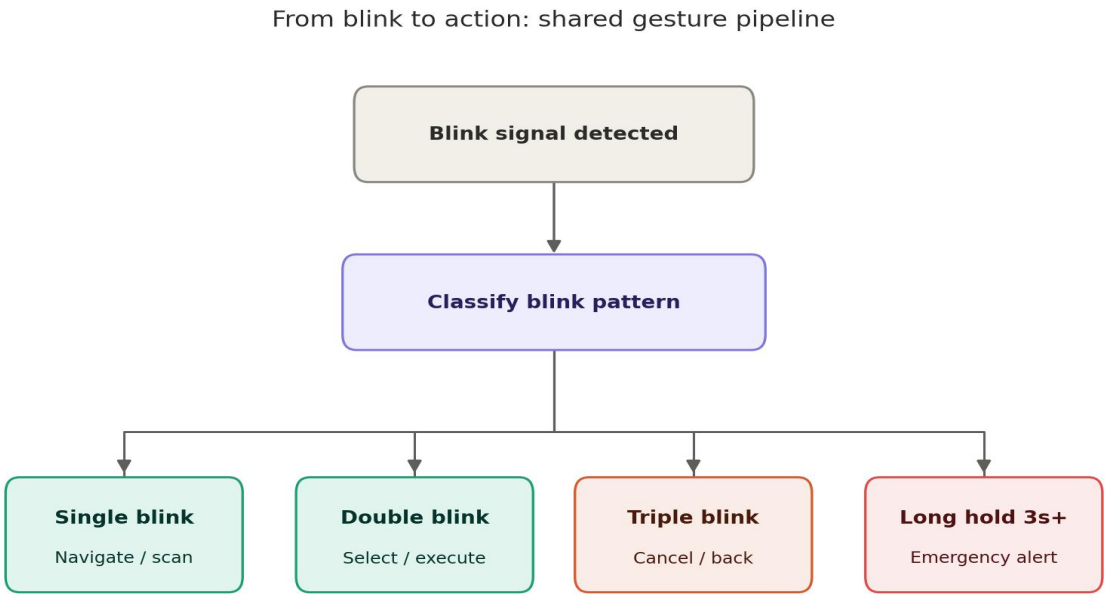
Electrooculography (EOG) makes this possible without a camera. When the eye moves or blinks, the electrical potential between the cornea and the retina shifts slightly, and electrodes placed around the eye can pick up that shift. Compared to camera-based eye-gaze systems, an EOG approach is lower cost, does not depend on lighting conditions, and keeps working even with the eyes closed - which is exactly what the emergency gesture in this project relies on.





Typical electrode placement for a 2-channel EOG setup.

Every use case in this project - communication, room control, physical actuation, and emergency alerting - runs through the same simple pipeline: detect a blink, classify its pattern, then route it to whatever mode the user is currently in. Reusing one gesture vocabulary across every mode means the user never has to relearn a new set of gestures when they switch tasks.



The shared gesture pipeline - one classification step feeding four possible actions.

Note that the emergency gesture sits apart from the others by design. It has to work even if everything else in the system fails, so it is built to interrupt whatever mode the user is currently in and to be effectively impossible to trigger by accident.

2. Gesture vocabulary

The system defines four gestures, each mapped to a distinct interface role:

Gesture	Interface role	Example use case
Single blink	Navigation (scan)	Moving to the next letter on a screen, selecting the next option in the main menu, or cycling to the next room appliance.
Double blink	Selection (execute)	Pressing "Enter", or toggling the selected switch on/off.
Triple blink	Cancel / back	Deleting a typed letter, or returning to the main menu.
Long hold (3s+)	Emergency override	Squeezing eyes shut to sound an alarm or call a caregiver.

3. Application use cases

3.1 Digital communication (AAC)

A simple web page hosted directly on the ESP32 displays a grid of letters or common phrases. The system automatically highlights rows one by one. A single blink skips ahead, and a double blink selects the highlighted letter to build a sentence, which the web browser can then read aloud.

3.2 Smart room automation

Direct control over the physical environment without needing a smartphone. Mapping a double blink to an ESP32 digital output pin can trigger a relay to switch high-power loads - turning on a desk lamp, activating a heater, or controlling a motorized fan.

3.3 Actuating physical mechanisms

Blinks translated into mechanical assistance. The ESP32's PWM capabilities can physically unlock a door latch or dispense medication from a hopper when a specific blink sequence is registered.

3.4 Emergency alerts

Safety features override all other modes. A "long hold" (eyes squeezed shut for three seconds) creates a large, sustained voltage spike that is effectively impossible to trigger by accident. This can be mapped to a loud buzzer, or use the ESP32's WiFi to send an automated alert (webhook to Telegram/WhatsApp, or a phone call) instantly notifying a caregiver.

All four modes can be combined into a single website, where the user selects which function they want to use at a given time.

4. Hardware and signal acquisition

4.1 Electrodes

Ag/AgCl electrodes are the standard choice for EOG, placed around the eyes as shown in Figure 1. A common layout uses electrodes placed laterally next to each eye for the horizontal channel, plus one above and below an eye for the vertical channel, with a reference electrode near the ear or cheek.

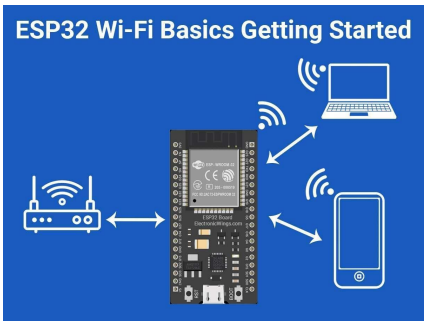
4.2 Amplification

The corneo-retinal potential is only a few microvolts to millivolts, far too small for a microcontroller's ADC to read directly, so it must be amplified first. An instrumentation amplifier - common choices include the INA118, INA121, or AD620/AD622 - provides high gain and a high common-mode rejection ratio (CMRR), which cancels out noise picked up equally by both electrodes. Gain is typically set with a single resistor and tuned to the 25x-500x range depending on electrode placement and signal strength.

4.3 Filtering

After amplification, a notch filter removes 50/60 Hz mains hum, and a cascade of low-pass filters smooths the signal. A DC bias correction (baseline restoration) stage is also needed, since the resting potential drifts over time as the eyes adapt to ambient light.

4.4 Digitizing



The conditioned analog signal feeds into the ESP32's built-in ADC. Because the ESP32's ADC can be noisy and nonlinear, budget time for calibration, or consider an external ADC such as the ADS1115 for cleaner readings.

5. Safety and patient isolation

- Power the sensing front-end from a battery, not mains - never let the amplifier circuit share a ground with anything plugged into a wall outlet.
- Use opto-isolated relays for any mains-powered actuator (lamp, heater) so there is no electrical path back to the user.
- Use skin-safe, medical-grade or surgical-tape-compatible electrodes, and plan for skin irritation during long-term wear.
- Treat the emergency channel as safety-critical: prioritize avoiding false negatives (a missed emergency signal) over avoiding false positives.

6. Blink classification algorithm

- Amplitude thresholding to detect that a blink occurred at all.
- A timing window (roughly 600 ms-1 s) to count how many blinks occurred in quick succession, distinguishing single, double, and triple blinks.
- A duration/hold check for the long-hold emergency gesture, separate from the blink counter.
- A per-user calibration routine (e.g. blink three times at startup) to set thresholds, since EOG signal strength varies from person to person.
- As a stretch goal, combining EOG with EMG data can help distinguish an intentional blink from a natural, involuntary one, since voluntary blinks tend to differ in duration and amplitude.

7. Hardware bill of materials

Component	Role
Ag/AgCl electrodes (3-5)	Pick up the small corneo-retinal signal from the skin around the eyes.
Instrumentation amplifier (INA118 / INA121 / AD622)	Amplifies the differential EOG signal; high CMRR rejects common-mode noise.
Op-amp filter stage (notch + low-pass)	Removes 50/60 Hz mains hum and smooths the signal before digitizing.
ESP32	Reads the conditioned signal (ADC), runs the classification logic, hosts the web UI, and handles WiFi.
Relay module (opto-isolated)	Switches mains-powered appliances (lamp, fan, heater)

Component	Role
	safely, without a shared ground.
Servo / actuator + driver	Drives physical mechanisms such as a door latch or medication dispenser.
Buzzer	Local audible alert for the emergency mode.
LiPo battery + charging module	Powers the sensing front-end in isolation from mains electricity.

8. Software and system architecture

- ESP32 hosts a local web page for the AAC letter/phrase grid, as well as pages for room control and actuation.
- A state machine handles mode switching between AAC, automation, actuation, and emergency - with emergency able to interrupt any other mode.
- Webhook integration sends alerts to Telegram/WhatsApp or triggers a phone call.
- Signal and event logging supports debugging and per-user calibration.

9. Further research directions

- Usability testing: even a small study with volunteers wearing electrodes helps validate that the system works outside a lab bench.
- Cost breakdown: a full bill-of-materials cost estimate strengthens the case that the system is realistically deploy-able.
- Comparison to existing solutions: eye-tracking cameras, sip-and-puff switches, and EMG switches - and why EOG is a good trade-off (cheap, no camera needed, works with eyes closed).
- Ethics and privacy: since alerts are sent over WiFi to a caregiver, briefly address data handling and privacy.

10. Circuit design in detail

This section walks through the analog front-end stage by stage, from the raw electrode signal to a value the ESP32 can read.

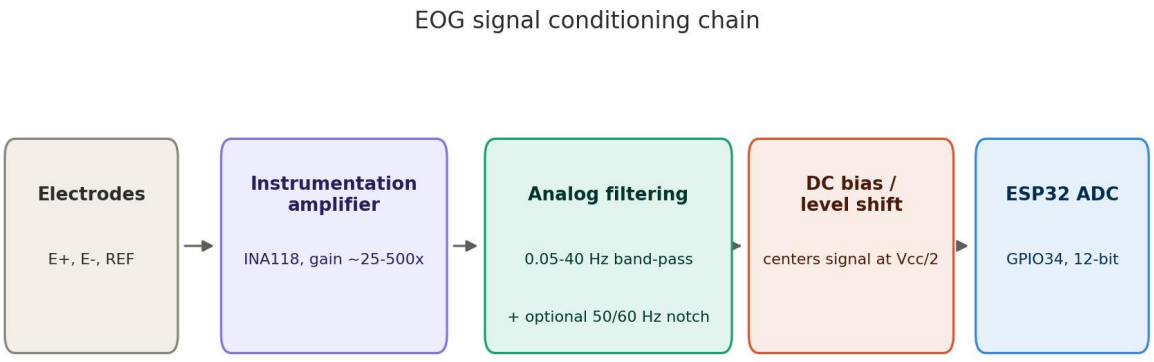


Figure 3: The full signal conditioning chain, from electrodes to the ESP32's ADC input.

5.1 Instrumentation amplifier stage

The corneo-retinal potential is only microvolts to a few millivolts, so the first job of the circuit is clean, high-gain differential amplification. An instrumentation amplifier such as the INA118 takes the two electrode inputs (E+ and E-) and amplifies only their difference, rejecting noise that appears equally on both (mains hum, muscle activity from elsewhere on the face, etc.).

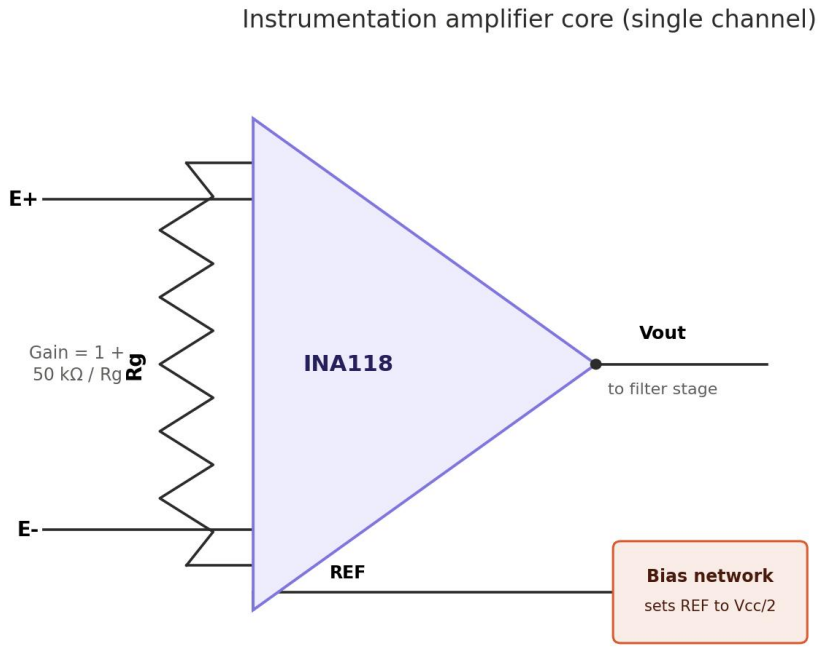


Figure 4: Instrumentation amplifier core - a single external resistor (Rg) sets the gain.

Gain is set with one external resistor, Rg, using the formula printed on the diagram: Gain = 1 + 50 kΩ / Rg. Two worked examples:

- Rg = 2 kΩ -> Gain = 1 + 50,000/2,000 = 26x - a moderate gain, good as a starting point during calibration.
- Rg = 100 Ω -> Gain = 1 + 50,000/100 = 501x - a high gain used once the signal is confirmed clean, to make small eye movements clearly visible.

The REF pin sets the amplifier's output reference level. Rather than tying REF straight to ground, it is tied to a bias network (see 5.3) so the amplified signal can swing both positive and negative around a safe midpoint instead of being clipped at 0V.

5.2 Filter stage

After amplification, two passive filters shape the signal:

- High-pass filter (~0.05 Hz cutoff): removes slow DC drift caused by the eyes adapting to ambient light, without removing the blink signal itself. A typical RC combination is R = 1 MΩ with C = 3.3 uF (fc = 1 / (2*pi*R*C) ≈ 0.048 Hz).

- Low-pass filter (~40 Hz cutoff): removes high-frequency muscle noise (EMG) while keeping blink energy, which is concentrated below about 10-20 Hz. A typical combination is R = 1.6 kOhm with C = 2.2 uF (fc ≈ 45 Hz).
- 50/60 Hz notch (optional): mains hum can also be removed here with an analog twin-T notch filter, but in practice it is often simpler and cheaper to apply a digital notch filter in the ESP32 firmware instead of adding analog components.

5.3 DC bias / level shift

The ESP32's ADC reads a uni polar 0-3.3V range, but the amplified EOG signal is an AC-coupled waveform that swings both above and below its resting level. To make full use of the ADC's range, a simple resistor-divider bias network (buffered by an op-amp voltage follower) sets a virtual ground at Vcc/2 (about 1.65V), and this becomes both the REF point for the instrumentation amplifier and the baseline the filtered signal rides on. Without this step, roughly half of every blink waveform would be clipped at 0V.

5.4 Powering the front-end

The entire sensing chain - electrodes, instrumentation amplifier, filters, and bias network - should run from an isolated battery supply rather than a USB or mains-derived rail, to avoid ever sharing a ground with mains electricity. A single 3.3V or 5V rail is sufficient; the bias network is what allows a single-supply circuit to still handle a signal that swings in both directions.